

$x \sim y$ if and only if $2|(x-y)$

$R: \{2, 4, 6, 8\}$

The relationship would be transitive and symmetric. If we accept that 2 can divide 0, then it would also be reflexive.

Reflexive:

$$x = 2, y = 2 \Rightarrow 2 - 2 = 0 \text{ if } 2 \mid 0$$

Symmetric:

$$x = 2, y = 4 \Rightarrow 2 \mid (2 - 4) \text{ and } 2 \mid (4 - 2)$$

Transitive:

$$x = 2, y = 4, z = 6$$

$$2 \mid (2 - 4), 2 \mid (4 - 6), \text{ and } 2 \mid (2 - 6)$$